

Implementation of a Simplified Deep Learning Pipeline for Nailfold Capillary Segmentation and Density Estimation

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Abstract—Nailfold capillaroscopy is a non-invasive imaging technique used to observe microvascular abnormalities that may support the diagnosis of systemic sclerosis and related vascular disorders. Manual assessment is often slow, subjective, and dependent on specialist expertise. This project implements a simplified deep learning workflow inspired by Bharathi *et al.*, who proposed an automated system for quantitative assessment of microvascular abnormalities in nailfold capillary images. Unlike the original clinical pipeline, which combines U-Net, ResNet34, morphological feature aggregation, and logistic regression, this implementation focuses on a student-level reproduction of the core technical idea: supervised U-Net segmentation followed by post-processing, quantitative measurement extraction, and density estimation. A professor-provided dataset containing segmentation image-mask pairs and density images with capillary-count annotations was used. The system successfully completed dataset preparation, training, validation, testing, mask generation, overlay visualization, measurement export, and density evaluation. The final segmentation accuracy was limited, with a mean Dice score of 0.0165 and mean IoU of 0.0084 on the held-out test set, while the density task achieved a mean absolute count error of 4.897 capillaries. Although the model is not clinically reliable, the work demonstrates an end-to-end biomedical image analysis workflow and provides a reproducible basis for future improvement.

Index Terms—Nailfold capillaroscopy, U-Net, deep learning, biomedical image segmentation, capillary density, medical image analysis, quantitative morphology.

I. INTRODUCTION

Nailfold capillaroscopy is used to visualize small blood vessels at the base of the fingernail. These capillary structures are clinically important because changes in width, density, and shape can be associated with microvascular disorders, especially systemic sclerosis (SSc) and Raynaud’s phenomenon [1]. In practice, capillaroscopy images are commonly interpreted by trained experts. However, manual interpretation can be time-consuming, subjective, and difficult to standardize across clinicians and hospitals.

Deep learning provides a possible solution because convolutional neural networks can automatically learn visual patterns from medical images. In the reference article, Bharathi *et al.* developed a multi-stage automated system that used U-Net for apex candidate generation and vessel segmentation,

ResNet34 for candidate classification and width estimation, and logistic regression for subject-level SSc probability prediction [1]. Their system achieved high diagnostic performance and showed that quantitative morphology features can support clinical interpretation.

The objective of this project was to implement a simplified version of this idea using the available coursework dataset. The goal was not to reproduce the complete clinical-grade system exactly. Instead, the project aimed to demonstrate the main implementation steps: preparing image-mask data, training a U-Net segmentation model, predicting binary masks, generating overlays, extracting simple vessel measurements, and evaluating capillary density using count annotations.

II. REFERENCE METHOD AND PROJECT SCOPE

A. Original Article Pipeline

The original article proposed an automated pipeline that converts nailfold capillary images into quantitative features and a subject-level SSc probability. Figure 1 shows the main article figure: the raw mosaic image, initial apex candidates, selected distal capillaries, and ROC performance curves. This figure is useful because it illustrates the difference between raw image acquisition, candidate detection, candidate refinement, and diagnostic evaluation.

B. Implemented Scope

The complete reference system includes several stages that were beyond the scope of this student implementation, especially ResNet34 candidate classification, width regression, patient-level feature aggregation, and logistic regression diagnosis. Therefore, this project focused on the parts that could be implemented and evaluated with the provided data: binary segmentation and density-related measurement. The implemented pipeline is shown in Fig. 2.

III. DATASET

The professor-provided dataset consisted of two folders. The first folder was used for segmentation and contained native nailfold images paired with binary masks. These masks acted as ground-truth labels for supervised U-Net training. The

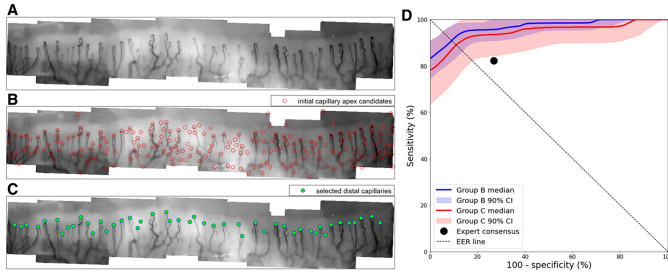


Fig. 1: Reference article visual example from Bharathi *et al.* [1]. Panel A shows the original nailfold mosaic, Panel B shows initial capillary apex candidates, Panel C shows selected distal capillaries after refinement, and Panel D shows ROC curves for diagnostic performance.

Simplified implementation workflow

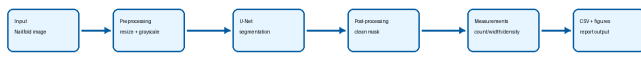


Fig. 2: Simplified implementation workflow used in this project. The system receives nailfold images, applies pre-processing, predicts segmentation masks using U-Net, post-processes the masks, extracts measurements, and exports figures and CSV files.

TABLE I: Prepared dataset summary.

Task	Train	Validation	Test/Eval
Segmentation	49	10	11
Density estimation	—	—	29

second folder was used for density estimation and contained native images together with versions that included a 1 mm scale. For this density task, the annotation was the number of capillaries per image.

The preparation script found 70 segmentation image-mask pairs. These were divided into 49 training images, 10 validation images, and 11 test images. The density task evaluated 29 images after one missing native image was skipped automatically by the script. Table I summarizes the prepared dataset. This separation was important because the test images were not used during training, making the final test results a more realistic measure of generalization.

IV. METHODOLOGY

A. Preprocessing

All segmentation images were loaded as grayscale images and resized to 256×256 pixels. Pixel intensities were normalized to the range $[0, 1]$. Binary masks were also resized to the same resolution and thresholded to obtain foreground and background labels. This preprocessing step ensured that all images had a consistent input size for the network.

Simplified U-Net architecture used in the implementation

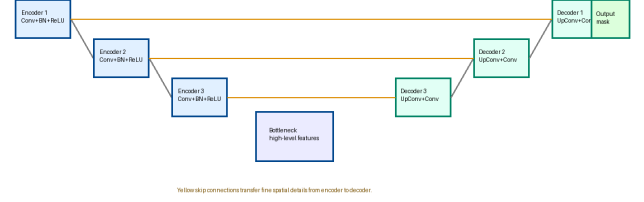


Fig. 3: Simplified U-Net architecture used in this project. The encoder extracts features, the bottleneck stores high-level compressed information, and the decoder reconstructs the segmentation mask. Skip connections preserve fine capillary details.

B. U-Net Architecture

U-Net was selected because it is one of the most widely used architectures for biomedical image segmentation [2]. It has an encoder-decoder structure with skip connections. The encoder extracts visual features while reducing spatial resolution. The bottleneck stores the most compressed high-level representation of the image. The decoder upsamples the features and reconstructs a pixel-level segmentation map. Skip connections copy fine spatial details from the encoder to the decoder, which is important for thin and small structures such as capillaries.

The implemented U-Net used convolution, batch normalization, and ReLU activation blocks. It produced a one-channel probability map indicating whether each pixel belonged to a vessel region. This probability map was converted to a binary mask using a threshold.

C. Training Strategy

Training was performed using supervised learning. For each training image, the network predicted a binary mask and compared it with the ground-truth mask. The loss function combined binary cross-entropy and Dice loss:

$$\mathcal{L} = \mathcal{L}_{BCE} + \mathcal{L}_{Dice}. \quad (1)$$

Binary cross-entropy penalizes wrong pixel classifications, while Dice loss encourages overlap between predicted masks and ground-truth masks. The model was trained for 20 epochs on CPU using the Adam optimizer. The best model was saved according to validation Dice score.

D. Post-Processing and Measurement Extraction

After prediction, the binary masks were cleaned using morphological operations. Small noisy components were removed. Connected components were then used as approximate vessel/capillary regions. From the cleaned mask, several simplified quantitative measurements were extracted:

- **Component count:** estimated number of detected vessel/capillary regions.

TABLE II: Comparison between the reference article and the implemented student version.

Aspect	Bharathi <i>et al.</i> reference system	Implemented student system
Main goal	Subject-level SSc probability from nailfold capillary images	Simplified segmentation and density estimation from provided images
Models	U-Net, ResNet34, width regression, feature aggregation, logistic regression	U-Net segmentation, morphological post-processing, connected-component counting
Labels used	Capillary apexes, vessel regions, morphology features, subject diagnosis	Binary segmentation masks and capillary-count annotations
Output	SSc diagnostic probability and quantitative morphology measurements	Predicted masks, overlays, CSV measurements, density/count error
Clinical readiness	Research-level clinical study with expert comparison	Educational implementation; not clinically reliable

TABLE III: Main implementation settings.

Setting	Value
Model	U-Net
Input size	256×256 pixels
Image format	Grayscale
Optimizer	Adam
Loss	BCE + Dice loss
Epochs	20
Device	CPU
Train/val/test split	49 / 10 / 11
Prediction threshold	0.5

- **Mean width:** average estimated vessel width calculated using distance transform and skeletonization.
- **Maximum width:** largest estimated vessel width in the image.
- **Density per 100 pixels:** number of components normalized by the horizontal image span.
- **Shape score:** approximate irregularity/tortuosity score derived from contour geometry.

These measurements are simplified approximations of the morphology features used in the original paper, such as width, shape, and density [1].

V. EXPERIMENTAL SETUP

The experiment was implemented in Python using PyTorch, OpenCV, NumPy, pandas, matplotlib, and scikit-learn. The program was executed on CPU. The complete script automatically prepared the data, trained the U-Net, generated predictions on the held-out test set, created overlays and binary masks, saved measurement CSV files, and evaluated the density task.

Table III summarizes the main implementation settings. These settings were selected to keep the experiment feasible on CPU while still demonstrating a complete deep learning workflow.

For segmentation, Dice score and Intersection over Union (IoU) were used. Dice and IoU measure the overlap between the predicted mask and the ground-truth mask. A higher value means better segmentation. They are defined as:

$$Dice = \frac{2|P \cap G|}{|P| + |G|}, \quad (2)$$

$$IoU = \frac{|P \cap G|}{|P \cup G|}, \quad (3)$$

where P is the predicted mask and G is the ground-truth mask. For density estimation, the predicted component count was compared with the provided capillary-count annotation. Mean absolute count error was used to summarize performance:

$$MAE = \frac{1}{N} \sum_{i=1}^N |\hat{c}_i - c_i|, \quad (4)$$

where $\hat{c}_i$ is the predicted count and c_i is the ground-truth count.

A. Implementation Procedure

The Python implementation followed the same practical order used in many biomedical imaging projects:

- 1) unzip and organize the segmentation and density folders;
- 2) match each native image with its corresponding mask;
- 3) split the segmentation data into training, validation, and test subsets;
- 4) train the U-Net using supervised image-mask learning;
- 5) save the best model based on validation Dice score;
- 6) generate predicted masks and green overlays for test images;
- 7) calculate Dice, IoU, component count, width, density, and shape score;
- 8) compare predicted density counts with the provided annotations;
- 9) export figures and CSV files for report writing and later GitHub submission.

This procedure makes the work reproducible because the same script generates both the trained model and all quantitative outputs used in this report.

VI. RESULTS

A. Segmentation Results

The segmentation experiment completed successfully, but the final accuracy was low. Table IV summarizes the main segmentation metrics. The mean Dice score was 0.0165 and the mean IoU was 0.0084. The best test Dice score was 0.0564. This indicates that the model produced masks and measurements, but the overlap with the provided ground-truth masks was weak.

B. Density Estimation Results

For density evaluation, predicted components were counted and compared with the provided ground-truth capillary counts. Table VI summarizes the results. The mean absolute count

TABLE IV: Segmentation result summary on the held-out test set.

Metric	Value
Number of test images	11
Mean Dice score	0.0165
Mean IoU	0.0084
Best Dice score	0.0564
Mean component count	6.36
Mean estimated width (px)	7.35

TABLE V: Example per-image segmentation outputs from the test set.

Image	Dice	IoU	Components
N23a.jpg	0.0242	0.0123	7
N25c.jpg	0.0189	0.0095	12
N29d.jpg	0.0334	0.0170	9
N30g.jpg	0.0317	0.0161	15
N31b.jpg	0.0564	0.0290	3

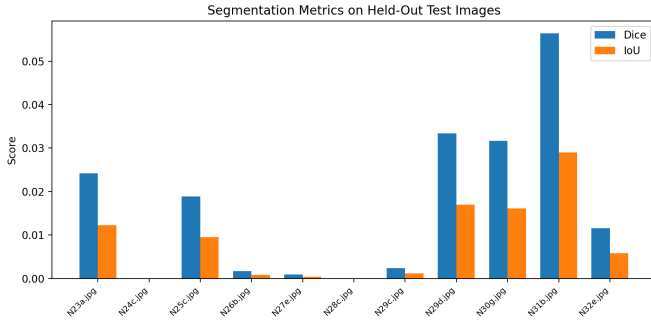


Fig. 4: Dice and IoU scores for each held-out segmentation test image. The values show that the model ran successfully but did not learn accurate segmentation boundaries.

TABLE VI: Density estimation summary.

Metric	Value
Evaluated images	29
Mean ground-truth count	5.45
Mean predicted count	5.66
Mean absolute count error	4.897

error was 4.897 capillaries. The mean ground-truth count was 5.45, while the mean mean predicted count was 5.66. Although the average count scale was close, the individual image errors remained large.

VII. DISCUSSION

This project successfully implemented an end-to-end biomedical image analysis workflow. The system prepared the dataset, trained a U-Net model, generated segmentation outputs, extracted measurements, and evaluated density estimation. This is important because it demonstrates the practical link between the article methodology and a real implementation.

However, the quantitative segmentation performance was poor. This does not mean the implementation failed; it means

the simplified model did not learn sufficiently accurate masks under the current experimental conditions. Several reasons can explain this result. First, the training set was small, containing only 49 training images. Second, training was performed on CPU for only 20 epochs. Third, the implemented model was intentionally lightweight for a student project. Fourth, nailfold capillaries are thin structures, so even small localization errors strongly reduce Dice and IoU scores. Finally, the original article did not rely on U-Net alone; it used a more complete system with U-Net, ResNet34, feature aggregation, and logistic regression [1].

The implementation is therefore best understood as a simplified reproduction of the technical workflow, not a clinical diagnostic system. The main achievement is not high accuracy, but successful implementation of the core stages: supervised learning with ground truth masks, segmentation prediction, post-processing, feature measurement, and density evaluation.

A. Comparison With the Original Article

Bharathi *et al.* achieved high diagnostic performance because their system used a multi-stage architecture. U-Net generated candidates, ResNet34 filtered and classified candidates, another ResNet34 estimated width, and logistic regression combined patient-level features into SSc probability [1]. In contrast, the present project only implemented U-Net segmentation and simplified measurement extraction. Therefore, lower performance is expected. This comparison is important because it shows why clinical-grade AI systems usually require more than a single segmentation model.

B. Strengths and Limitations

The main strength of this project is that it is reproducible and complete. It uses real image-mask data, trains a model, produces output masks, creates overlays, exports CSV measurements, and evaluates performance. The results are also reported honestly, including weak Dice and IoU scores.

The limitations are also clear. The dataset is small, no GPU was used, only a basic U-Net was implemented, and the density estimation relies on connected components that may not correspond exactly to true capillaries. The model also lacks the ResNet34 refinement stage used in the reference article.

C. Future Work

Future improvements should include training for more epochs, using GPU acceleration, adding data augmentation, tuning the prediction threshold, improving post-processing, and adding a ResNet34 classifier to remove false positives. A stronger version could also combine extracted features using logistic regression to reproduce the article's final diagnostic stage. For future biomedical imaging work, the same workflow can be extended to MRI tumour segmentation by replacing nailfold images with MRI scans and replacing vessel masks with tumour masks.

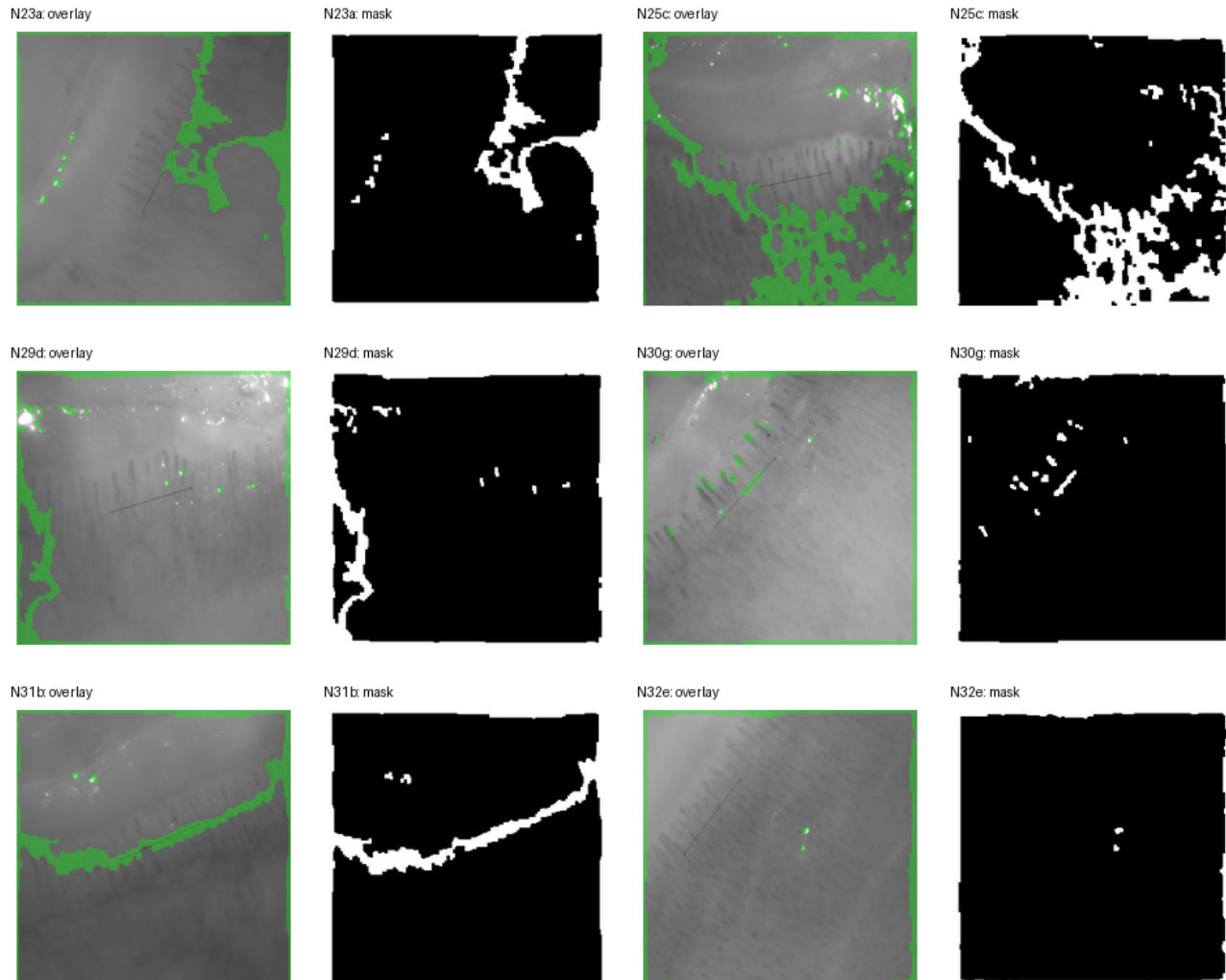


Fig. 5: Examples of U-Net segmentation outputs on held-out test images. Each pair shows the predicted overlay and binary mask. These examples are useful for qualitative interpretation of the model behavior.

VIII. REPRODUCIBILITY AND GITHUB ORGANIZATION

This project is available as a reproducible student biomedical AI implementation on GitHub:

<https://github.com/petricks001/nailfold-capillary-deep-learning>

The repository contains the Python implementation, report, requirements file, generated figures, segmentation outputs, and reproducible workflow scripts used in this project. The raw medical datasets are not intended for public redistribution unless permission is granted by the course instructor. Therefore, the repository mainly focuses on the implementation pipeline, generated outputs, and non-sensitive examples.

The implemented workflow includes:

- dataset preparation,
- supervised U-Net training,
- segmentation prediction,

- overlay generation,
- quantitative measurement extraction,
- density evaluation,
- CSV export of results.

The repository structure is organized as follows:

```
nailfold-capillary-deep-learning/
nailfold_student_implementation.py
README.md
requirements_nailfold.txt
report.pdf

nailfold_work/
  prepared/
  outputs/
  raw/
```

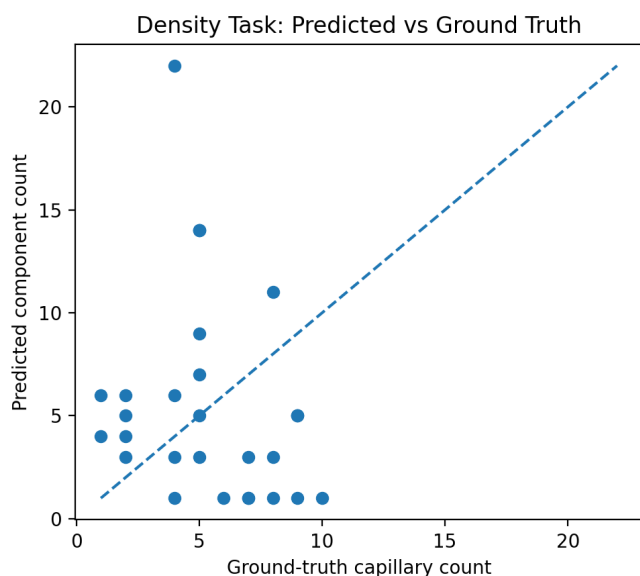


Fig. 6: Predicted component count versus ground-truth capillary count for the density task. Points near the dashed diagonal line represent more accurate predictions.

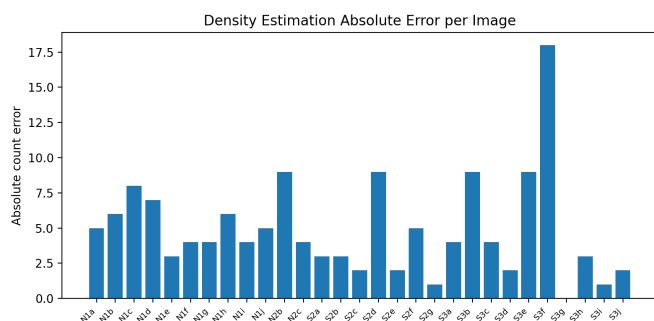


Fig. 7: Absolute capillary-count error per density image. Some images have large errors, showing that the simplified segmentation-based counting method needs improvement.

This organization makes the implementation reproducible and useful for future biomedical imaging projects, including more advanced applications such as MRI tumour segmentation using image-mask supervised learning workflows.

IX. CONCLUSION

This project implemented a simplified deep learning system for nailfold capillary segmentation and density estimation. A U-Net model was trained using professor-provided image-mask pairs, and the trained model was used to generate predicted masks, overlays, and quantitative measurements. The implementation produced a complete experimental workflow, including dataset preparation, model training, test prediction, CSV export, and density evaluation. Although the segmentation accuracy was low, the project demonstrates the practical foundation of biomedical AI: image-mask supervised learning,

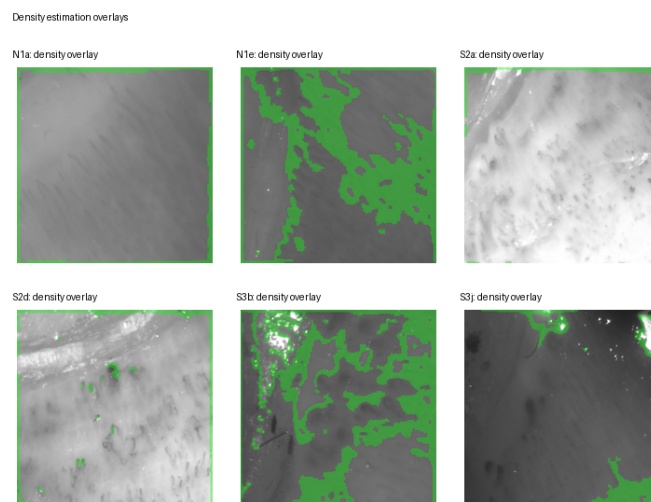


Fig. 8: Example density overlays generated by the implementation. The highlighted regions were used to estimate component counts and approximate density measurements.

segmentation, measurement extraction, and critical interpretation of results. The work provides a strong base for future improvements and for related medical imaging tasks such as MRI tumour segmentation.

ACKNOWLEDGMENT

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